

Solutions to Problems

30.

(a) 3-Methylbutanoyl iodide;

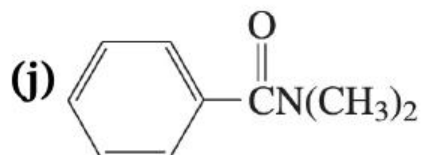
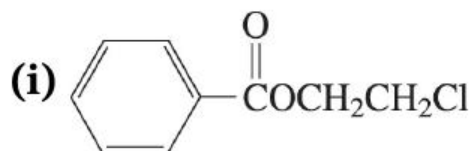
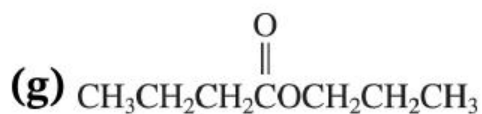
(b) 1-methylcyclopentanecarbonyl chloride;

(c) 2,2,2-trifluoroacetic anhydride;

(d) benzoic propanoic anhydride;

(e) ethyl 2,2-dimethylpropanoate;

(f) *N*-phenylacetamide





31.

(a) Methyl 2-chloro-3-butenate or methyl 2-chlorobut-3-enoate;

(b) ethyl 1-hydroxycyclobutanecarboxylate;

(c) *N*-methyl-2-oxobutanamide;

(d) 5-formyl-1,3-benzenedicarboxamide.

32. The guiding principle is that the strength of an acid is related to the ability of its conjugate base to accommodate the negative charge that remains after loss of a proton ([Section 2-2](#)). We are given ([Section 20-1](#)) that the order of acidity of α -hydrogens in carboxylic acid derivatives increases in the series carboxamide (weakest acid) < ester < anhydride < acyl halide < acyl chloride (strongest acid). (a) We know ([Section 18-1](#)) that resonance delocalization of the negative charge of an enolate from the α -carbon into the carbonyl group is partly responsible for the acidity of the α -hydrogen. Resonance donation of an electron pair

from L in $\text{R}\overset{\text{O}}{\parallel}\text{CL}$ to the carbonyl group essentially competes with

possible donation from the deprotonated α -carbon. ~~α -carbon~~. The net effect is destabilization of the anionic enolate. Therefore, as the resonance donation ability of L increases, the acidity of the α -hydrogen ~~α -hydrogen~~ decreases. **(b)** Inductively, as the electronegativity of L increases, stabilization of the negative charge of the enolate function should increase accordingly, as is observed.

33.

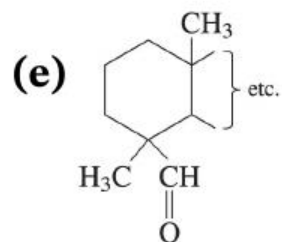
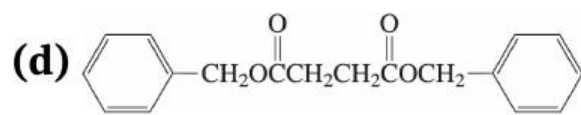
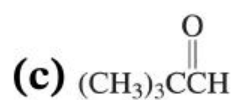
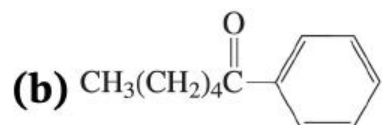
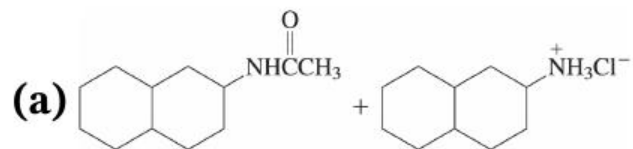
(a) Acetyl chloride (ClCl is bigger than F,F , and the bonds to it are longer);

(b) $\text{CH}_2(\text{COCH}_3)_2$ $\text{CH}_2(\text{COCH}_3)_2$ (F's F's $\alpha\alpha$ to ketones are more acidic than H's H's $\alpha\alpha$ to esters);

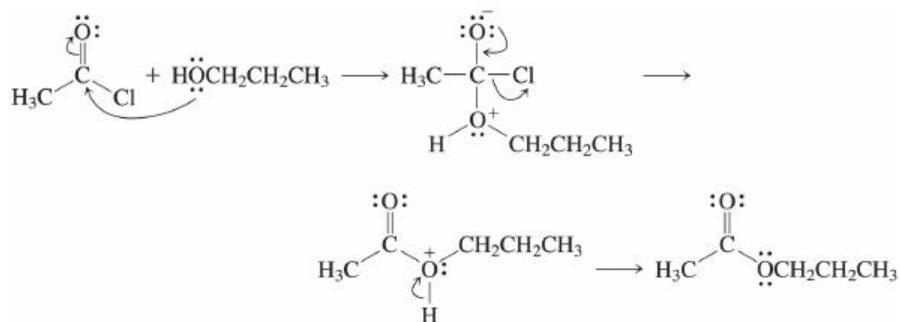
(c) Imide (The lone pair on NN is shared in resonance by two carbonyl groups, so it does not reduce their electrophilicity as much as the NN in an amide. Note that the relationship between an imide and an amide is similar to that between an anhydride and an ester.)

(d) Ethenyl acetate $\text{CH}_3\text{C}(=\text{O})-\ddot{\text{O}}-\text{CH}=\text{CH}_2 \longleftrightarrow \text{CH}_3\text{C}(=\text{O})-\ddot{\text{O}}^+-\text{CH}=\ddot{\text{C}}\text{H}_2$
 resonance **reduces** electron donation from oxygen toward the carbonyl carbon. So the resonance form $\text{CH}_3\text{C}(\text{O}^-)=\text{O}^+-\text{CH}=\text{CH}_2$ is reduced in importance, strengthening the $\text{C}-\text{OC}-\text{O}$ double bond and raising the carbonyl stretching frequency, actually to about 1760 cm^{-1} .)

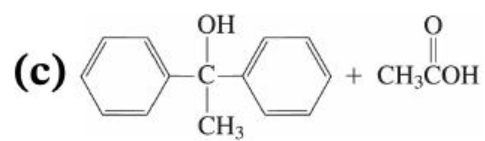
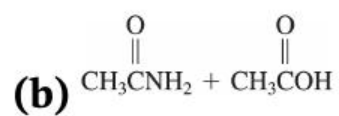
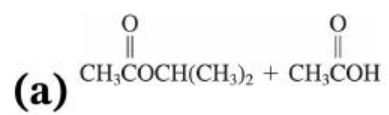
34.



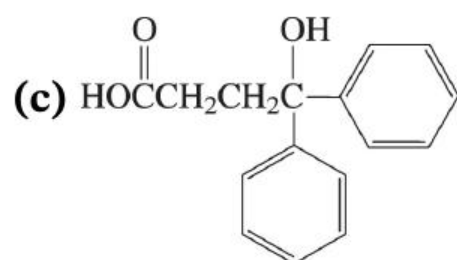
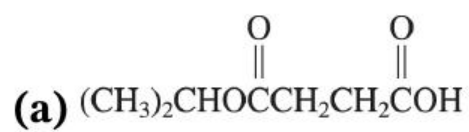
35.



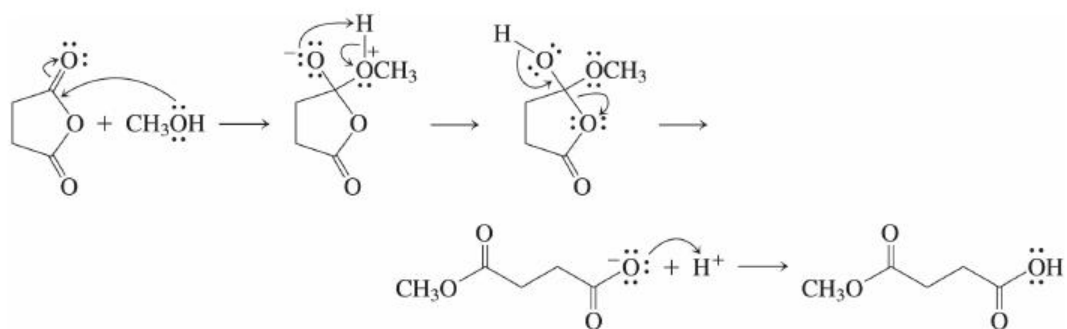
36. Assume aqueous acid work-up for both this and the next problem.



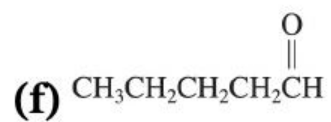
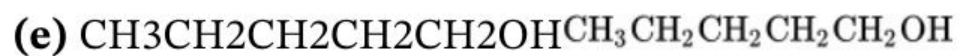
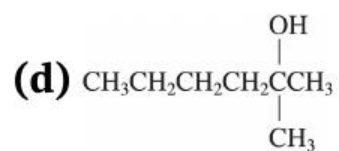
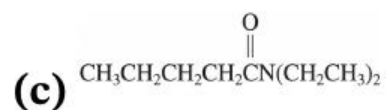
37.



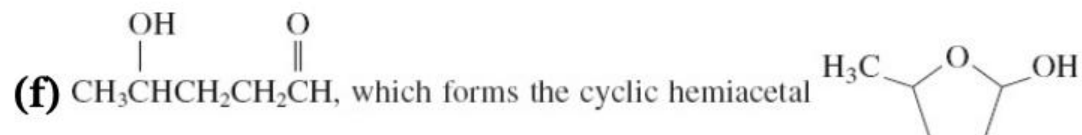
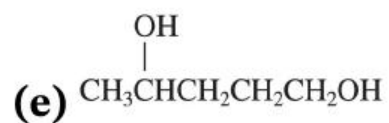
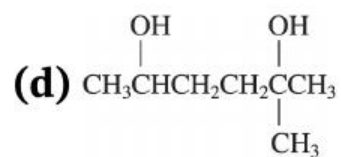
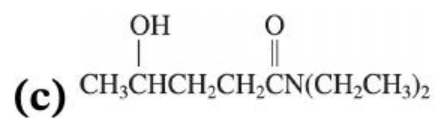
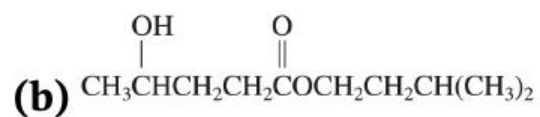
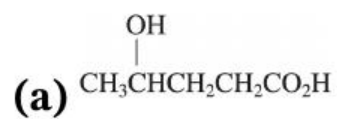
38.



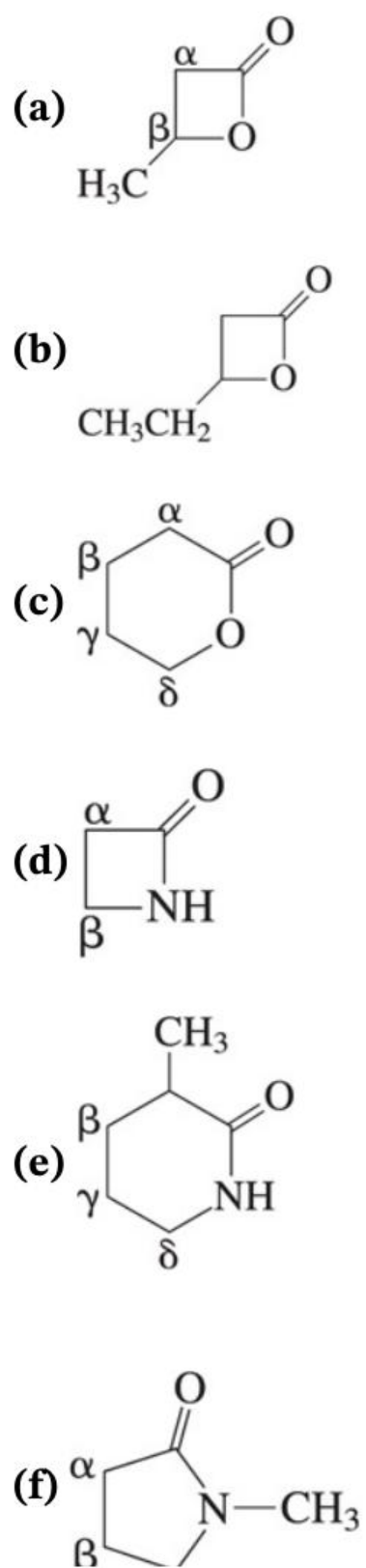
39.



10.

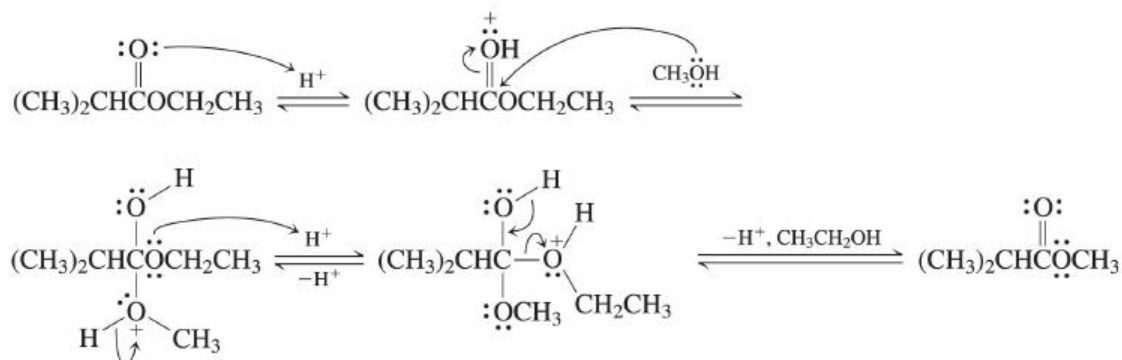


7.1.

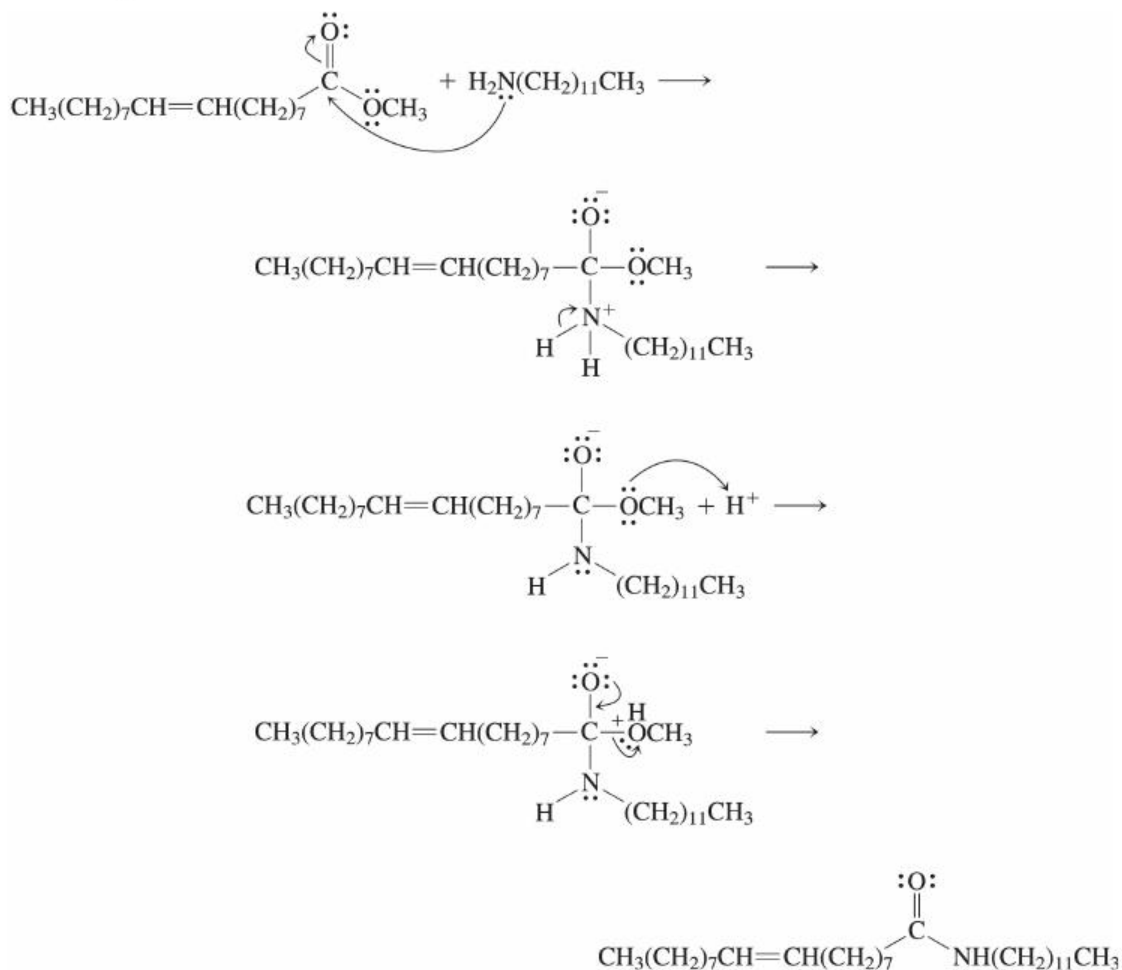


~ γ

12.



13.



14.

(a) Hexanoic acid;

(b) CH_3NH_2 , CH_3NH_2 , heat;

(c) DIBAL, -60°C ; -60°C ;

(d) H^+ or $-\text{OH}$, H_2O $-\text{OH}$, H_2O heat;

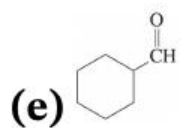
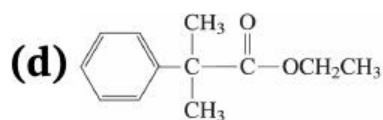
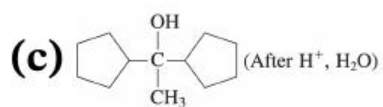
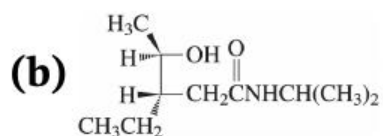
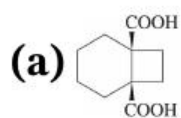
(e) LiAlH_4 ; LiAlH_4 ;

(f) Br_2 , NaOH , H_2O Br_2 , NaOH , H_2O

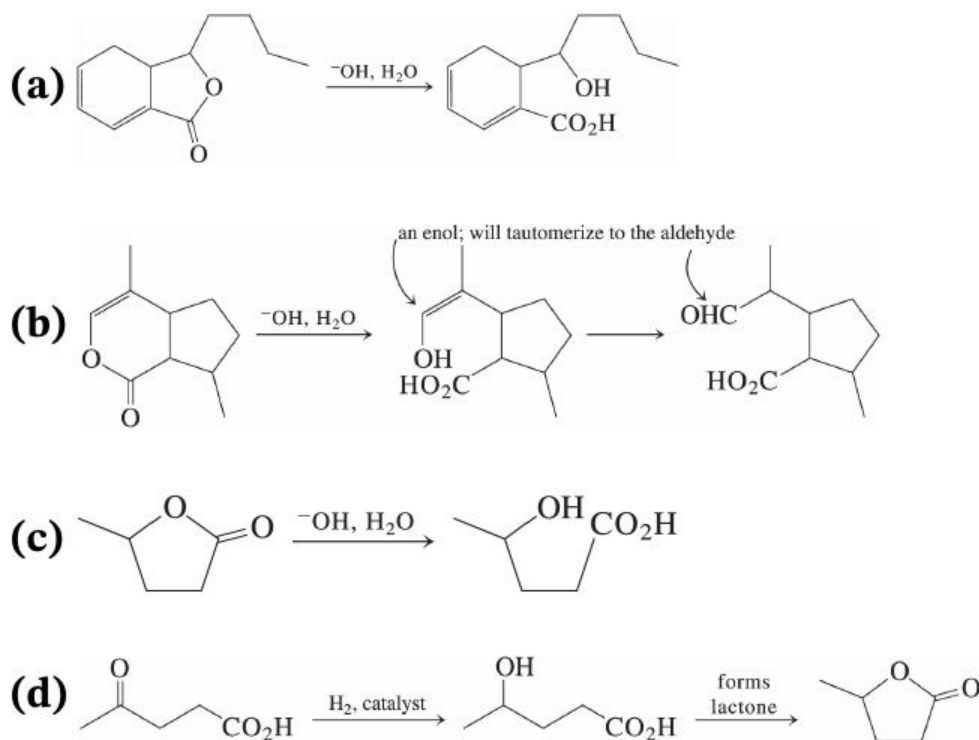
(g) $\text{CH}_3\text{CH}_2\text{MgBr}$ $\text{CH}_3\text{CH}_2\text{MgBr}$ (2 equivalents);

(h) $\text{C}_6\text{H}_5\text{MgBr}$. $\text{C}_6\text{H}_5\text{MgBr}$.

15.

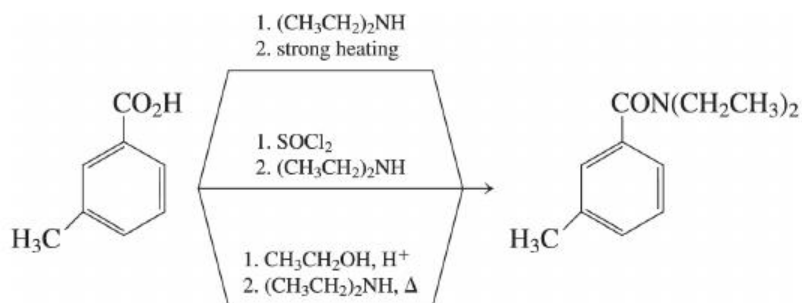


16. Each of the first three answers assumes subsequent acidification to convert the carboxylate salt into the neutral acid.



17.

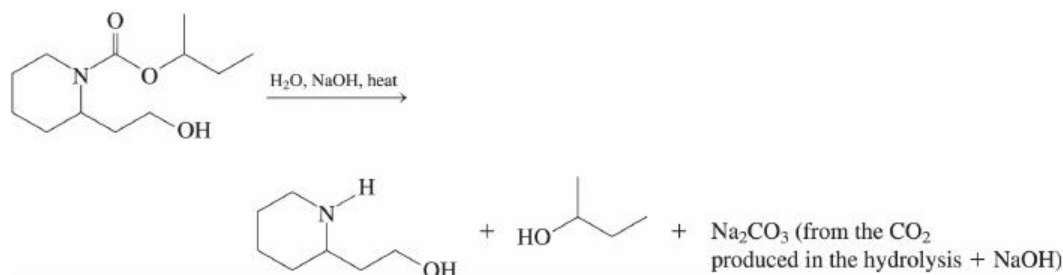
(a) Amides may be made directly from the acids, or indirectly via acyl halides, anhydrides (less efficiently, because only one of the two acyl groups is converted into an amide), or esters. Thus we have the following options:



Method 1: Direct reaction between an amine and an acid first gives a salt; strong heating converts the salt to the amide with loss

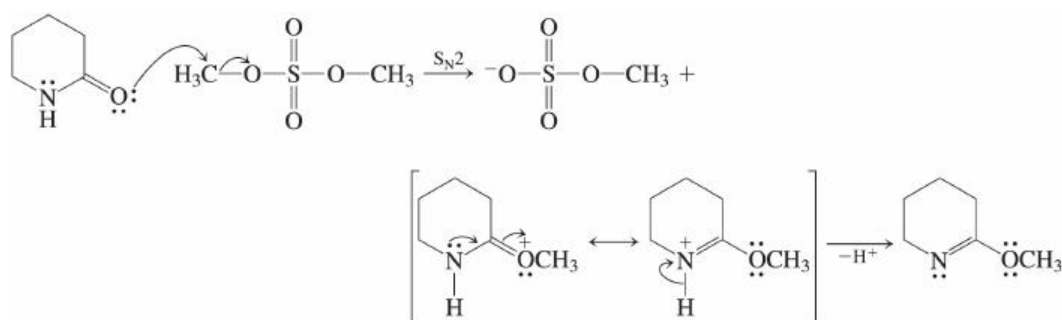
of water. Method 2: Converts to acyl chloride; reaction with amine gives amide with loss of HCl. Often done in the presence of excess amine to consume the HCl. Method 3: Esterification with any inexpensive alcohol (ethanol chosen here), then reaction with the amine under mild heating.

- (b) The group is an example of a carbamate or urethane, an ester of a carbamic acid (text p. 990). With lone pairs from *both* an oxygen and a nitrogen capable of donating electron density into the carbonyl group, it would be expected to be less reactive than any of the common derivatives discussed in the chapter.
- (c) Hydrolysis should proceed as shown below, cleaving both the amide and ester parts of the urethane function.

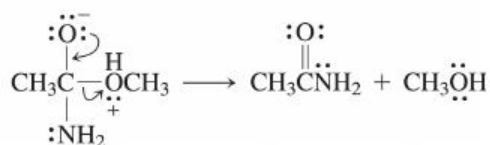
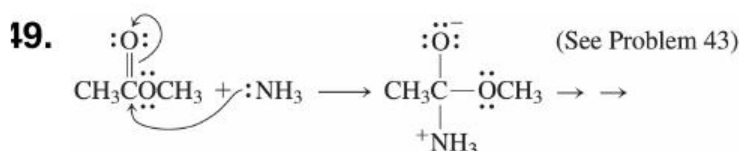


18.

- (a) Dimethyl sulfate is a highly reactive S_N2 substrate with respect to attack by a nucleophile on a methyl group (similar to CH₃I in reactivity). The process displaces the methyl sulfate anion, a good leaving group. The starting cyclic amide (a lactam) possesses two possible nucleophilic atoms, nitrogen and oxygen. Reaction of the oxygen with an electrophile gives a resonance-stabilized intermediate; reaction of the nitrogen does not. Therefore, we can proceed as follows:

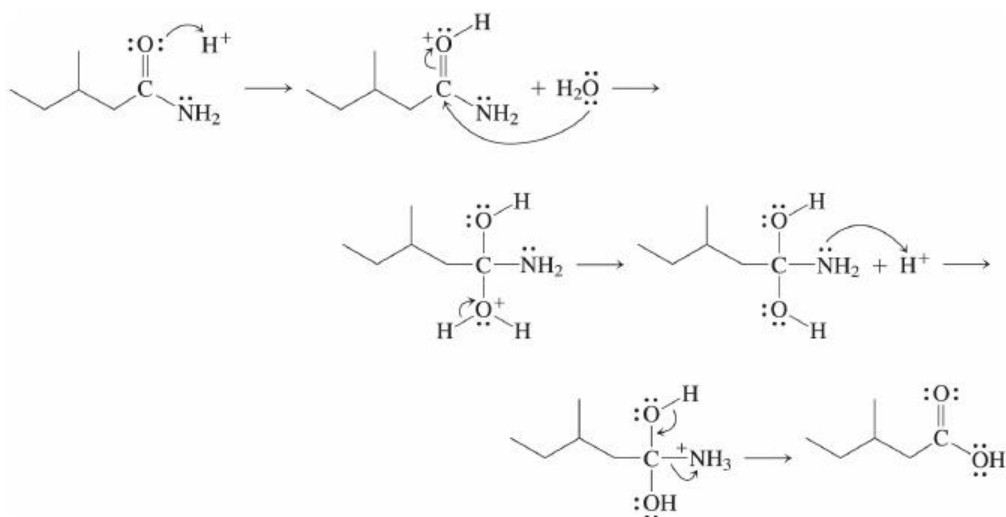


- (b) Ritalin has two stereocenters, one being the substituted carbon in the ring, the other the attached carbon in the side chain. Therefore, it will have four stereoisomers, two pairs of enantiomers that have diastereomeric relationships with each other. The active stereoisomer has the *SS* configuration at the ring carbon and is *RR* at the side-chain stereocenter.



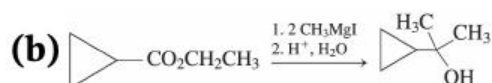
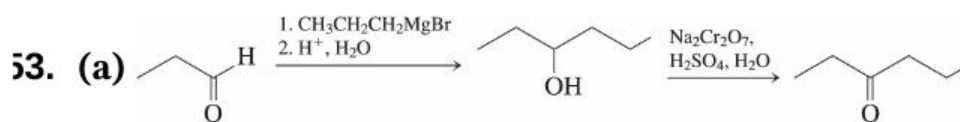
50. Pentanamide gives (a) pentanoic acid, (e) pentanamine, and (f) pentanal; N,N-*N,N*-dimethylpentanamide gives the same products for (a) and (f), whereas LiAlH_4 reduction in this case gives N,N-*N,N*-dimethylpentanamine, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{N}(\text{CH}_3)_2$.

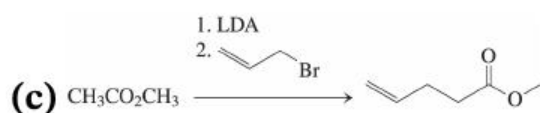
51.



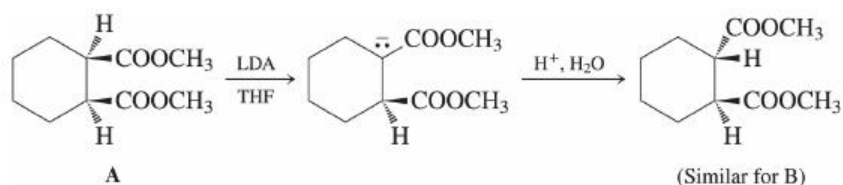
52.

- (a) $(\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2)_2\text{CuLi}$; $(\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2)_2\text{CuLi}$; then H^+ , H_2O ; H^+ , H_2O ;
- (b) (d) LiAlH_4 , $(\text{CH}_3\text{CH}_2)_2\text{O}$; LiAlH_4 , $(\text{CH}_3\text{CH}_2)_2\text{O}$; then H^+ , H_2O ; H^+ , H_2O ;
- (c) $\text{LiAl}[\text{OC}(\text{CH}_3)_3]\text{H}$; $\text{LiAl}[\text{OC}(\text{CH}_3)_3]\text{H}$; then H^+ , H_2O ; H^+ , H_2O ;
- (e) (f) $\text{CH}_3\text{CH}_2\text{CH}_2\text{MgBr}$, $(\text{CH}_3\text{CH}_2)_2\text{O}$; $\text{CH}_3\text{CH}_2\text{CH}_2\text{MgBr}$, $(\text{CH}_3\text{CH}_2)_2\text{O}$; then H^+ , H_2O ; H^+ , H_2O



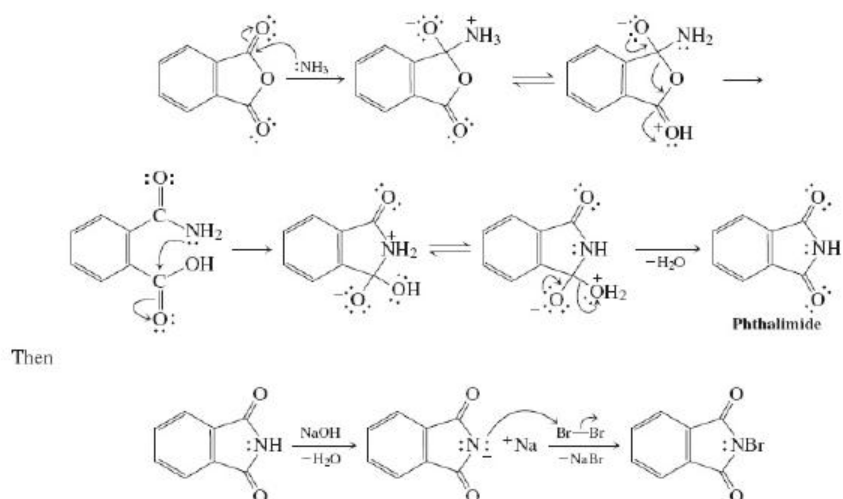


54. In A and B, the most acidic hydrogens are α -hydrogens. α -hydrogens. Deprotonation followed by protonation provides a mechanism for isomerization.

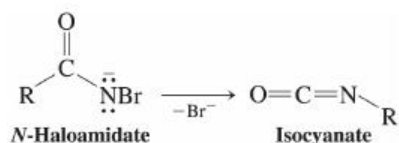


In C the most acidic hydrogens are on nitrogen. The α -hydrogens are not removed, so no isomerization is observed.

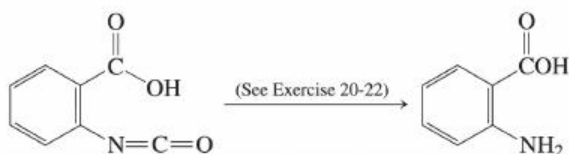
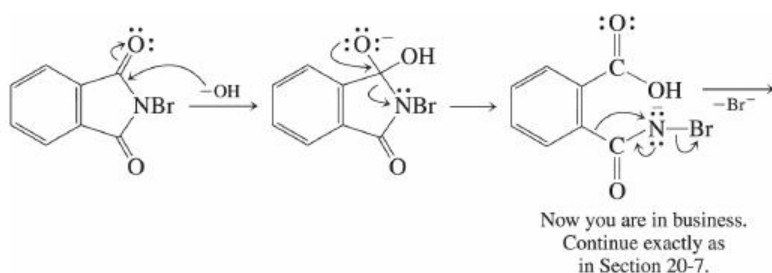
55. The mechanisms are related to amide formation and Hofmann rearrangement, respectively. In the formation of phthalimide, only those proton transfers that occur along the pathway to the product are shown.



Now what? There's no hydrogen on the NN anymore, so how can you get to an intermediate N-N-haloamidate that can rearrange to the isocyanate?

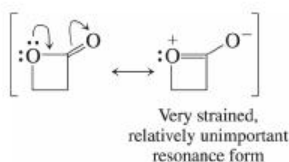


The answer: The reaction takes place in **strong base**, so use hydroxide in addition-elimination.



56. Self-explanatory.
57. From A: 1. SOCl_2 , SOCl_2 , 2. $(\text{CH}_3)_2\text{NH}$, $(\text{CH}_3)_2\text{NH}$ (makes carboxamide), 3. LiAlH_4 , $(\text{CH}_3\text{CH}_2)_2\text{O}$, LiAlH_4 , $(\text{CH}_3\text{CH}_2)_2\text{O}$, then H^+ , H_2O , H_2O From B: 1. SOCl_2 , SOCl_2 , 2. NH_3 , NH_3 , 3. Cl_2 , NaOH , Cl_2 , NaOH (Hofmann rearrangement, loses CO_2 and makes simple amine), 4. 2 CH_3I , NaOH , 2 CH_3I , NaOH ($\text{S}_\text{N}2$ methylations of amine nitrogen). Notice that B had an extra carbon that the Hofmann rearrangement removed.
58. The dipolar resonance forms that weaken the carbonyl $\text{C}=\text{O}$ bond and reduce its stretching frequency are less and less favorable in small rings because of the increased strain associated with a second

sp^2 atom.

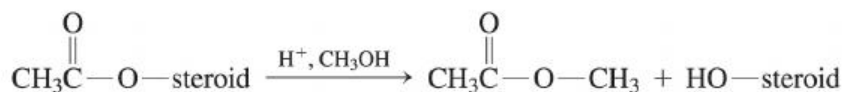


59.

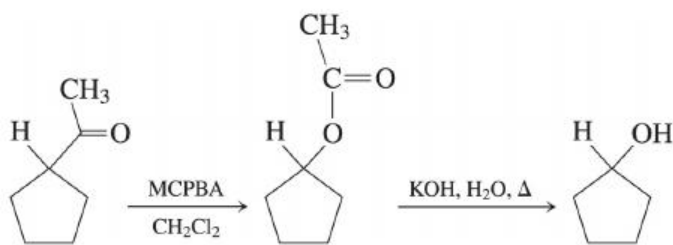
(a) Bad idea. Residual hexanoyl chloride will be converted to hexanoic acid, which smells like a herd of goats on a hot day.

(b) Wash out the glassware with an alcohol like ethanol. Reaction with hexanoyl chloride produces the ester ethyl hexanoate, which smells like fresh fruit. Much better.

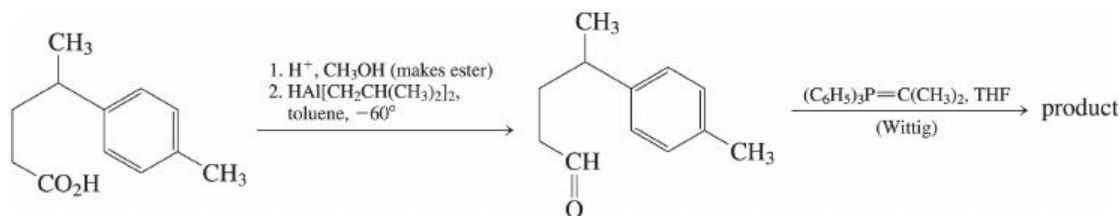
30. React with H^+ and CH_3OH ! The methyl ester group at the upper right will remain unchanged because the only nucleophile around for attacking carbonyl groups is methanol. However, the acetate function at the lower right will exchange to give methyl acetate, and the steroid alcohol group will be displaced.



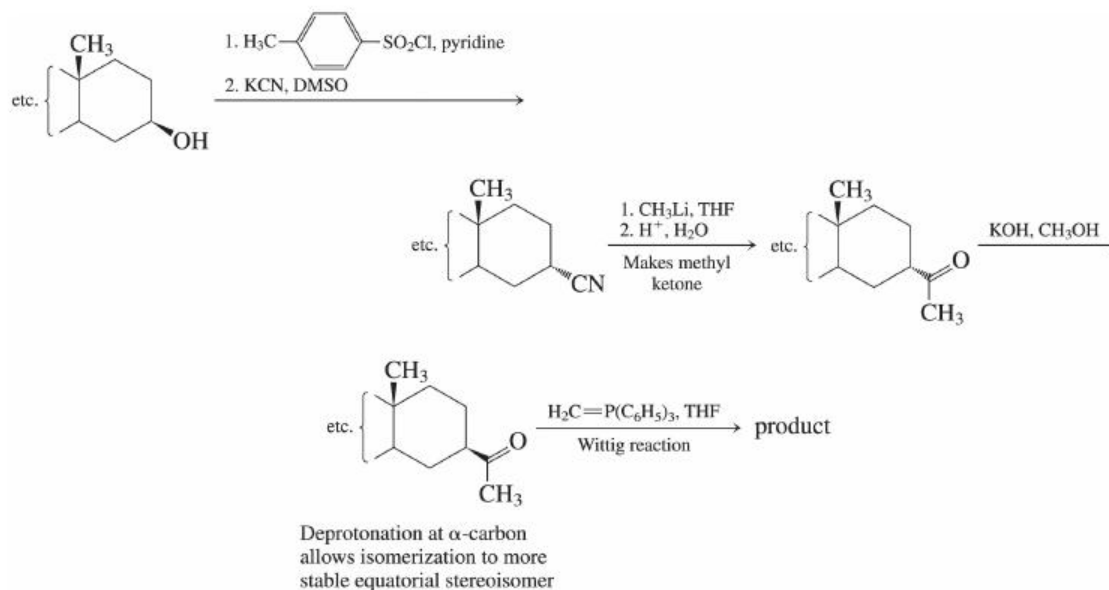
31. Baeyer-Villiger reaction, followed by ester hydrolysis.

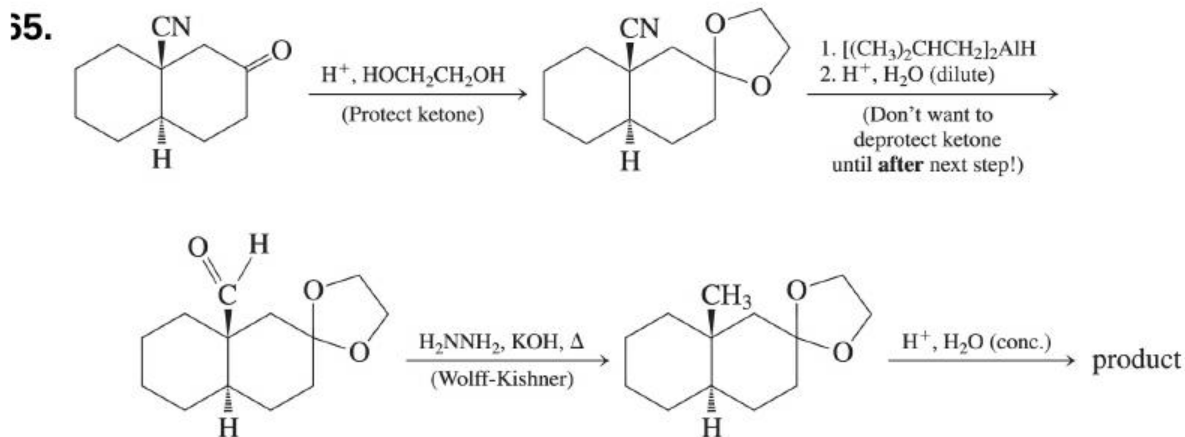


32. Several ways will work. One that uses a reaction from this chapter (the actual route used) is shown below.



33. 1. $(\text{CH}_3)_2\text{NH}$, Δ ($\text{CH}_3)_2\text{NH}$, Δ (makes *N,N,N,N*-dimethyl carboxamide),
2. LiAlH_4 , $(\text{CH}_3\text{CH}_2)_2\text{O}$ (makes amine)
 LiAlH_4 , $(\text{CH}_3\text{CH}_2)_2\text{O}$ (makes amine)
3. H^+ , H_2O
34. Pay attention to the stereochemistry. The solution shown is from the literature. A feasible alternative would be to set the desired stereochemistry of the nitrile in the beginning, e.g. 2. NaI , DMSO , then NaI , DMSO , then 3. KCN , DMSO .





36. (a) Collect your information and decide what it tells you. Begin with the molecular ion. It is a single peak, ruling out ClCl or Br₂, which would give two peaks two mass units apart. The value is an even number, therefore one N atom cannot be present (see Exercise 11-24 on text [page 505](#); zero or an even number of nitrogen atoms is required for an even mass number). The IR absorption is closest to the ester range, so you should operate under the initial assumption that the molecule is an ester and contains only C, H, and O.

The exact mass will give you the molecular formula, with a bit of work. The ester function contains two oxygens, so subtract their mass (2×15.9949) from that of the parent ion and try to match the remainder with the mass of some combination of carbons and hydrogens:

$$\begin{array}{r}
 116.0837 \\
 - (2 \times 15.9949) \\
 \hline
 116.0837 - (2 \times 15.9949) = 84.0939
 \end{array}$$

The only reasonable combination of C and H that has a mass of 84.0939 is C₆H₁₂. Does it match **exactly**?

$$(6 \times 12) + (12 \times 1.00783) = 84.0939$$

$(6 \times 12) + (12 \times 1.00783) = 84.0940$ —yes, indeed it does. (If that had not been the case, you would have had to explore possibilities containing oxygen, such as C_5H_8O , C_5H_8O , etc.) The molecular formula is $C_6H_{12}O_2$, $C_6H_{12}O_2$, giving one degree of unsaturation, the ester carbonyl group.

Now turn to the proton NMR spectrum. You have the following absorptions:

$\delta=1.1$: $\delta = 1.1$: Triplet, integration of 3 H; a H_3C-CH_2- group
3 H; a H_3C-CH_2- group

$\delta=1.2$: $\delta = 1.2$: Doublet, integration of 6 H; a $(CH_3)_2CH-$ group
6 H; a $(CH_3)_2CH-$ group

$\delta=2.3$: $\delta = 2.3$: Quartet, integration of 2 H; indicates a H_3C-CH_2- group, 2 H; indicates a H_3C-CH_2- group, possibly attached to a carbonyl carbon due to the slightly deshielded position in the spectrum

$\delta=5.0$: $\delta = 5.0$: A septet (see amplified scan), integration of 1 H; 1 H; seven lines imply two methyl groups as neighbors; chemical shift indicates attachment to oxygen: a $(CH_3)_2CH-O-(CH_3)_2CH-O-$ group

Do you have your answer? Yes: $CH_3CH_2C(=O)OCH(CH_3)_2$

Does this fit the rest of the data? The upfield region of the NMR spectrum can now be interpreted as consisting of a large doublet for the methyls of the $CH(CH_3)_2CH(CH_3)_2$ group, overlapping one peak of a smaller triplet, which arises from the CH_3CH_3 of the ethyl group. The carbon-13 NMR spectrum confirms these assignments. How about the mass spectrum? The base peak at $m/z=57$ arises from a type of α cleavage at the ester $C-OC-O$ bond, giving the very stable acylium ion $CH_3CH_2C\equiv O^+$. $CH_3CH_2C\equiv O^+$. See how many of the other fragments you can assign

fragments you can assign.

- (b) In this case you see two parent ions, two mass units apart, of equal intensity; a bromine atom is present. About half of the molecules contain ^{79}Br and a molecular weight of 180; the other half contains ^{81}Br with total $m/z=182$. $m/z = 182$. Again, the IR spectrum tells you that an ester is present.

Using either exact mass, subtract the mass of the appropriate Br isotope and that of the two oxygens of the ester function:

$$\begin{array}{r}
 179.9886 \\
 -78.9183 \\
 \hline
 101.0703 \\
 -(2 \times 5.9949) \\
 \hline
 69.0805
 \end{array}$$

179.9886-78.9183-101.0703-(2 × 5.9949)=69.0805

The only reasonable combination of C and H that has a mass of 69 is C_5H_9 . It has the correct exact mass:

$$(5 \times 12) + (9 \times 1.00783) = 69.0705. (5 \times 12) + (9 \times 1.00783) = 69.0705.$$

The molecular formula is $\text{C}_5\text{H}_9\text{O}_2\text{Br}$, giving one degree of unsaturation, the ester carbonyl group.

You could turn directly to the NMR data, but first try to extract all you can from the mass spectral data. The base peak has $m/z=29$; two possibilities you have seen in this chapter are CH_3CH_2^+ and $\text{HC}\equiv\text{O}^+$. A pair of peaks at $m/z=107$ and 109 imply a Br-containing fragment. What could it be? If you subtract the atomic mass of the appropriate Br isotope from either, you get 28, which could correspond either to C_2H_4 or CO. You can therefore write possible structures for the 107/109 fragment such as CH_3CHBr^+ and $\text{BrC}\equiv\text{O}^+$. The IR told you that you were dealing with an ester, not an acyl halide, so the first of these two possibilities is more plausible. You are getting very close. So look at the proton NMR spectrum. You have the

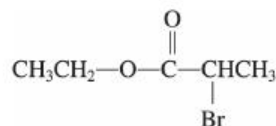
following absorptions:

$\delta=1.3$: $\delta = 1.3$: Triplet, integration of 3 H; a $\text{H}_3\text{C}-\text{CH}_2-$ group
3 H; a $\text{H}_3\text{C}-\text{CH}_2-$ group

$\delta=1.8$: $\delta = 1.8$: Doublet, integration of 3 H; a $\text{H}_3\text{C}-\text{CH}-$ group
3 H; a $\text{H}_3\text{C}-\text{CH}-$ group

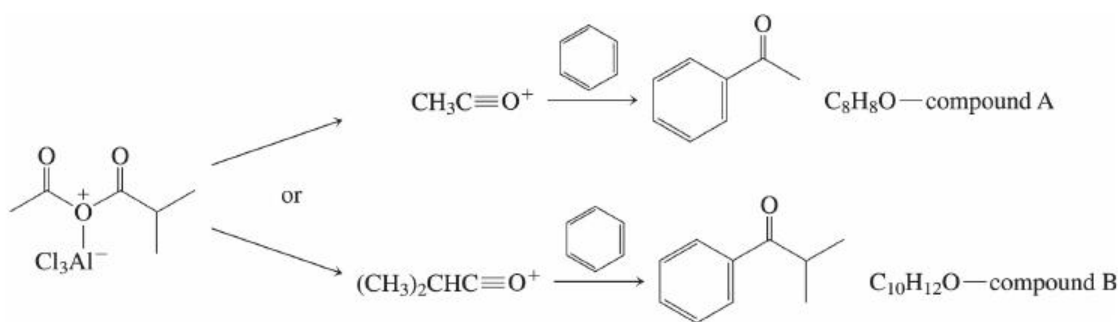
$\delta=4.2$: $\delta = 4.2$ and 4.4 : Quartets, with integrations of 2 H, 2 H and 1 H, 1 H, respectively.

These must be the CH_2 and CH groups inferred from the two signals, above! Because these protons are so deshielded, you can assume that one of these groups is attached to the ester oxygen and the other to bromine. Which is attached to which? If you try connecting the $\text{H}_3\text{C}-\text{CH}_2-$ to bromine, you make $\text{CH}_3\text{CH}_2\text{Br}$, bromoethane. That's no good because there's no place left to attach all the rest of the atoms. So, instead connect $\text{H}_3\text{C}-\text{CH}_2-$ to the ester oxygen and $\text{H}_3\text{C}-\text{CH}-$ to bromine, giving the answer

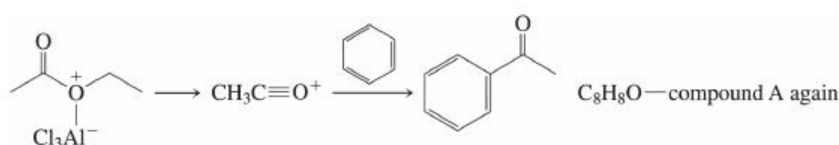


Again, the ^{13}C NMR spectrum is in agreement with these assignments.

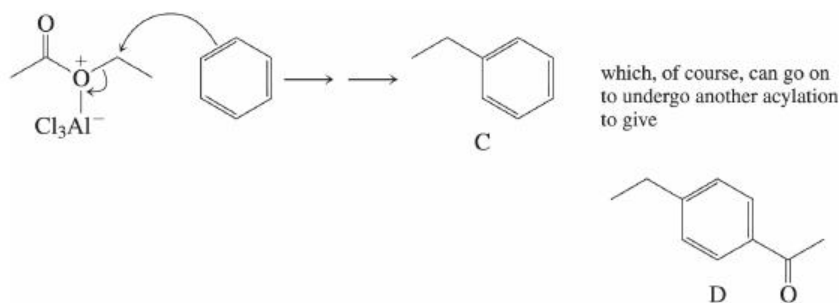
57. First reaction: The “mixed” anhydride can release either of two different acylium ions.



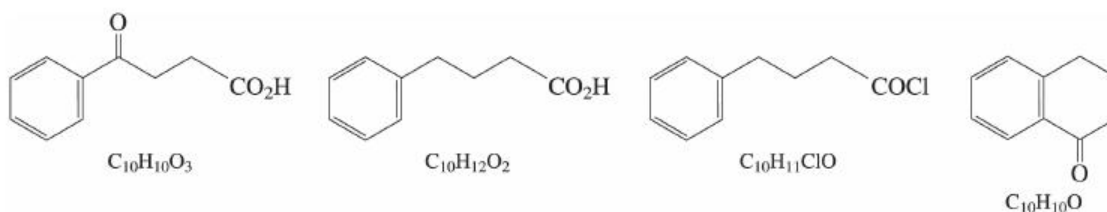
Second reaction: An ester can serve as a source of acylium ion, too.



But what else can happen? What is this product C, C_8H_{10} ? Look at the NMR: Five benzene hydrogens, and CH_2 and CH_3 groups upfield—looks like ethylbenzene! How? The Lewis acid complexed oxygen is a good leaving group, so we can propose the following:



Third reaction sequence:



38. (e)

69. (a)

70. (d)

71. (b)

21

Amines and Their Derivatives: Functional Groups Containing Nitrogen

Amines represent the last of the simple functional groups that you will encounter in organic chemistry. They are not entirely new to you, of course. They popped up as early as [Chapter 6](#), as the results of nucleophilic substitution reactions between ammonia and haloalkanes. More recently, you have been introduced to amine syntheses starting from carboxamides and nitriles ([Sections 20-6](#) through [20-8](#)). As usual, the chapter begins by presenting the usual body of descriptive information concerning the properties of amines as a class of compounds. It then focuses in more detail on the limitations and variations associated with several amine syntheses, finishing with sections on their reactions. Amines are of substantial importance biologically. But unlike other biologically important compound classes, amines are not involved in a very wide range of distinctly different types of reactions. Thus, you should find this aspect of amine chemistry relatively manageable.

Outline of the Chapter

[21-1 Nomenclature](#)

[21-2, 21-3, 21-4 Physical, Spectroscopic, and Acid-Base
Properties of Amines](#)

Qualitative and quantitative characteristics of the functional group.

21-5, 21-6, 21-7 Synthesis

Guidelines for choosing synthetic strategies.

21-8 through 21-10 Reactions

Mostly extensions of earlier material; a couple of special reactions, too.

Keys to the Chapter

21-1. Nomenclature

To an even greater extent than with most other compounds, common names for amines are still almost universally used. It is therefore necessary to be able to recognize structures from either their alkylamine- or aniline-based name. The systematic alkanamine method looks tricky at first but becomes simple once you recognize that it works in much the same way that the IUPAC alcohol (alkanol) naming system works.

21-2 through 21-4. Properties of Amines

Amines are related to alcohols in the same way that ammonia is related to water. This parallel makes the properties of amines easy to predict, because the main qualitative difference is simply that N is less electronegative than O. So, hydrogen bonding in amines is present, but it is weaker than in alcohols; deshielding of nearby ^1H and ^{13}C NMR signals is observed but to a lesser extent than in alcohols; IR spectra are similar; mass spectra are rather predictable. Tertiary amines (R_3N) may be viewed as nitrogen analogs of ethers (R_2O).

By the way, all these nice, tidy analogies do not extend to the **smells** possessed by most amines. Whereas alcohols tend to have, at worst, somewhat heavy, sweetish odors, amines, at best, smell like ammonia, and, at worst, richly deserve the common names that have been bestowed upon some of their representatives. These include names like cadaverine, putrescine, and skatole. Dead fish would be an improvement.

The acid-base properties of amines are an extension of what you know about ammonia: They are weaker acids and stronger bases than are water or alcohols. It will repay you many times over, however, to go over the information involving pK_a values of these molecules. The qualitative

ability to handle acid-base concepts is one of the more useful capabilities you can take out of a course in organic chemistry.

21-5, 21-6, and 21-7. Synthesis

By the time you finish reading these sections of the text, you will be aware of the fact that the first amine synthesis you learned, alkylation of NH_3 via SN_2 reaction, is also generally the worst amine synthesis. It is much better to use any one of the special N-containing nucleophiles that give clean monoalkylation products, which can then be turned into amines. For simple systems lacking sensitive functional groups, all the methods will work comparably well. The choice becomes more critical if the molecule is more sensitive. For example, 1. N_3^- , 2. LiAlH_4 would be a poor choice for the conversion of $\text{Br}(\text{CH}_2)_3\text{COCH}_3$ into the corresponding amine, because the ketone will be reduced by the hydride reagent along with the azide group. The ketone could be protected before starting, but a better solution would be the Gabriel sequence, which involves hydrolysis instead of reduction in the second step.

An important additional amine synthesis in this section is reductive amination of aldehydes and ketones. In particular, reductive amination of a ketone is a better way to make an amine attached to a 2° alkyl group than is SN_2 reaction with a 2° haloalkane. Reductive amination is also an important way by which nature makes amines.

You've already seen how to make an amine with one more carbon by using SN_2 reaction with CN^- , followed by reduction. What would you do, however, if you faced the particularly nasty problem of needing to make an amine attached to a 3° alkyl group (R_3CNH_2)? That's tricky, because SN_2 reactions won't work. If you could add a carbon somehow to get to R_3CCONH_2 , then you would be all set to use one of the above rearrangements, right? O.K., think about that one for awhile. You will find help in [Section 19-6](#). The solution will be given in the answer to [Problem 40\(b\)](#).

21-8 through 21-10. Reactions

Beyond simple displacement reactions in which an amine behaves as a nucleophile, there is a small group of specialized reactions of amines, each of which has a very specific use. The Hofmann elimination has been more important in structure determination than anything else. It is frequently combined with the Mannich reaction, however, to make an important synthetic entry to cyclic carbonyl compounds with methylene groups next to the carbonyl (see [Problem 52](#)). Such structures are present in many naturally occurring (plant-derived) antitumor agents.

The nitrosation reactions and the chemistry of diazoalkanes are also used in synthesizing just certain specific types of compounds, like cyclopropanes. The mechanisms here are more involved, however, and an understanding of their steps at this point is helpful because similar types of chemistry will be seen again later, in [Chapter 22](#). As you plow through this, try to focus on the relationship of each mechanistic step to processes you've seen before. Almost all of this chemistry is based on relatively fundamental sorts of events, like protonation and deprotonation, combined with elimination or addition.